Lecture 11 Feb 18 2026

7.2 trigonometric integrals

Example. Evaluate $\int \cos^3 x\, dx$.

Sln.

$$\cos^3 x = \cos x \cdot \cos^2 x = \cos x (1 - \sin^2 x) = \cos x - \cos x \sin^2 x.$$

then

$$\int \cos^3 x\, dx = \int (\cos x - \cos x \sin^2 x)\, dx$$

$$= \int \cos x\, dx - \int \cos x \sin^2 x\, dx \qquad u = \sin x,\ du = \cos x\, dx$$

$$= \sin x - \int u^2\, du$$

$$= \sin x - \frac{u^3}{3} + C$$

$$= \sin x - \frac{\sin^3 x}{3} + C.$$

Example. Find $\int \sin^5 x \cos^2 x\, dx$.

Sln. $\sin^5 x \cos^2 x = (\sin^2 x)^2 \cdot \sin x \cos^2 x.$ $\qquad \sin^2 x = 1 - \cos^2 x.$

$$= (1 - \cos^2 x)^2 \cos^2 x \sin x. \qquad u = \cos x,\ du = -\sin x\, dx.$$

$$\int \sin^5 x \cos^2 x\, dx = \int (1 - \cos^2 x)^2 \cos^2 x \sin x\, dx$$

$$= -\int (1 - u^2)^2 u^2\, du$$

$$= -\int (1 - 2u^2 + u^4) u^2\, du$$

$$= -\int (u^2 - 2u^4 + u^6)\, du$$

$$= -\frac{u^3}{3} - \frac{2}{5}u^5 + \frac{1}{7}u^7 + C$$

$$= -\frac{1}{3}\cos^3 x - \frac{2}{5}\cos^5 x + \frac{1}{7}\cos^7 x + C.$$

$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ $\qquad \cos^2 x = \frac{1}{2}(1 + \cos 2x).$

Example. Evaluate $\int_0^{\pi} \sin^2 x\, dx$.

Sln.

$$\int_0^{\pi} \sin^2 x\, dx = \frac{1}{2}\int_0^{\pi} (1 - \cos 2x)\, dx = \frac{1}{2}\pi - \frac{1}{4}\sin 2x \Big|_0^{\pi} = \frac{\pi}{2}.$$

$$= \frac{1}{2}\pi - \frac{1}{4}(\sin 2\pi - \sin 0)$$

$$\int \cos(\alpha x)\,dx = \frac{1}{\alpha}\sin\alpha x + C \qquad \int \sin\beta x\,dx = -\frac{1}{\beta}\cos\beta x + C.$$

Example. Find $\int \sin^4 x\,dx$

Sln.

$$\int \sin^4 x\,dx = \int (\sin^2 x)^2\,dx$$

$$= \int \left(\frac{1}{2}(1-\cos 2x)\right)^2 dx \qquad (x\cdot y)^n = x^n y^n$$

$$= \frac{1}{4}\int (1 - 2\cos 2x + \cos^2 2x)\,dx$$

$$= \frac{1}{4}\int \left(1 - 2\cos 2x + \frac{1}{2}(1+\sin 2x)\right)dx$$

$$= \frac{1}{4}\int \left(\frac{3}{2} - 2\cos 2x + \frac{1}{2}\sin 2x\right)dx$$

$$= \frac{3}{8}x - \frac{1}{2}\int \cos 2x\,dx + \frac{1}{8}\int \sin 2x\,dx$$

$$= \frac{3}{8}x - \frac{1}{2}\cdot\frac{1}{2}\sin 2x - \frac{1}{8}\cdot\frac{1}{2}\cdot\cos 2x + C$$

$$= \frac{3}{8}x - \frac{1}{4}\sin 2x - \frac{1}{16}\cos 2x + C.$$

Thm. (Strategy for evaluating $\int \sin^m x \cos^n x\,dx$.

a). If the power of cosine is odd ($n = 2k+1$), save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of sine:

$$\int \sin^m x \cos^{2k+1}(x)\,dx = \int \sin^m x \left(\cos^2 x\right)^k \cdot \cos x\,dx.$$

$$= \int \sin^m x \left(1-\sin^2 x\right)^k \cdot \cos x\,dx$$

$u = \sin x \Rightarrow$ the integral becomes in an integral of some polynomial.

$$= \int u^m (1-u^2)^k\,du.$$

b). If the power of sine is odd ($m = 2k+1$), save one factor and use $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of cosine.

$$\int \sin^{2k+1} x \cos^n x \, dx = \int (\sin^2 x)^k \cdot \cos^n x \cdot \sin x \, dx.$$

$$= \int (1 - \cos^2 x)^k \cos^n x \cdot \sin x \, dx$$

$u = \cos x \quad du = -\sin x$

$$= -\int (1 - u^2)^k u^n \cdot du.$$

c). If the powers of both sine and cosine are even, use the half-angle identity

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \qquad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

Some times

$$\sin x \cos x = \frac{1}{2} \sin 2x.$$

Integrals of Powers of Secant and Tangent.

Example. Evaluate $\int \tan^6 x \sec^4 x \, dx$.

Sln. $\sec^2 x = 1 + \tan^2 x$.

$$\int \tan^6 x \sec^4 x \, dx = \int \tan^6 x \sec^2 x \cdot \sec^2 x \, dx.$$

$$= \int \tan^6 x (1 + \tan^2 x) \cdot \sec^2 x \, dx.$$

$u = \tan x$

$du = \sec^2 x \, dx$

$$= \int u^6 (1 + u^2) \, du.$$

$$= \int u^6 \, du + \int u^8 \, du$$

$$= \frac{1}{7} u^7 + \frac{1}{9} u^9 + C$$

$$= \frac{1}{7} \tan^7 x + \frac{1}{9} \tan^9 x + C.$$

Example: Find $\int \tan^5\theta \sec^7\theta \, d\theta$.

Sln. $\int \tan^5\theta \sec^7\theta \, d\theta = \int \tan^4\theta \sec^6\theta \sec\theta \tan\theta \, d\theta$

$= \int (\tan^2\theta)^2 \sec^6\theta \cdot \sec\theta \tan\theta \, d\theta$

$\tan^2\theta = \sec^2\theta - 1 \longrightarrow = \int (\sec^2\theta - 1)^2 \sec^6\theta \cdot \underbrace{\sec\theta \cdot \tan\theta}\, d\theta$, $u = \sec\theta$, $du = \sec\theta\tan\theta \, d\theta$.

$= \int (u^2-1)^2 u^6 \, du.$

$= \int (u^4 - 2u^2 - 1) u^6 \, du$

$= \int (u^{10} - 2u^8 - u^6) \, du$

$= \frac{1}{11}u^{11} - \frac{2}{9}u^9 - \frac{1}{7}u^7 + C$

$= \frac{1}{11}\sec^{11}\theta - \frac{2}{9}\sec^9\theta - \frac{1}{7}\sec^7\theta + C.$

thm (Strategy for evaluating $\int \tan^m x \sec^n x \, dx$

a). If the power of secant is even ($n = 2k$) save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$.

$$\int \tan^m x \sec^{2k} x \, dx = \int \tan^m x \cdot (\sec^2 x)^k \, dx$$

$$= \int \tan^m x \, (\sec^2 x)^{k-1} \cdot \sec^2 x \, dx$$

$u = \tan x$

$du = \sec^2 x \, dx$

$\sec^2 x = 1 + u^2$

$$= \int u^m (1+u^2)^{k-1} \, du.$$

b). If the power of tangent is odd ($m = 2k+1$) save a factor of $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to express the remaining factors in terms of $\sec x$.

$$\int \tan^{2k+1} x \sec^n x \, dx = \int (\tan^2 x)^k \sec^{n-1} x \cdot \sec x \tan x \, dx$$

$$= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x \, dx$$

$u = \sec x$

$du = \tan x \sec x \, dx$

$$= \int (u^2 - 1)^k u^{n-1} \, du.$$

Thm: $\int \sec x \, dx = \ln|\sec x + \tan x| + C.$

Pf.

$$\int \sec x \, dx = \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx.$$

let $u = \sec x + \tan x$ $\quad du = (\sec x \tan x + \sec^2 x) \, dx$

$$= \int \frac{1}{u} \, du.$$

$$= \ln|u| + C.$$

$$= \ln|\sec x + \tan x| + C.$$

For you. Prove that $\int \csc x \, dx = \ln|\csc x - \cot x| + C.$

Example: $\int \tan^3 x \, dx$

Sln: $\int \tan^3 x \, dx = \int \tan x \cdot \tan^2 x \, dx = \int \tan x (\sec^2 x - 1) \, dx$

$$= \int \tan x \sec^2 x \, dx - \int \tan x \, dx.$$

$$= \int \sec x \cdot \sec x \tan x \, dx - \int \tan x \, dx$$

$$= \frac{1}{2} \sec^2 x - \int \tan x \, dx$$

$$= \frac{1}{2} \sec^2 x - \ln|\sec x| + C.$$

Example. Find $\int \sec^3 x \, dx$

Sln. $u = \sec x \qquad dv = \sec^2 x \, dx$

$du = \sec x \tan x \qquad v = \tan x$

By IP.

$$\int \sec^3 x \, dx = \sec x \tan x - \int \sec x \cdot \tan^2 x \, dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

Implies

$$2 \int \sec^3 x \, dx = \sec x \tan x + \int \sec x \, dx$$

$$\int \sec^3 x = \frac{1}{2} (\sec x \tan x + \ln|\sec x + \tan x|) + C.$$

Example. Evaluate $\int \sin 4x \cos 5x \, dx$

Sln. $\sin A \cos B = \frac{1}{2} [\sin(A-B) + \sin(A+B)]$

$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$

$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$

$A = 4x \qquad B = 5x$

$$\int \sin 4x \cos 5x \, dx = \frac{1}{2} \int (\sin(-x) + \sin(9x)) \, dx$$

$$= -\frac{1}{2} \int \sin x \, dx + \frac{1}{2} \int \sin(9x) \, dx$$

$$= \frac{1}{2}\cos x - \frac{1}{18}\cos 9x + C.$$

$\int \cot^m x \csc^n x\,dx$ using $1+\cot^2 x = \csc^2 x$.